

RAY-FINNED FISH

These fish are bony, with a hard, calcified skeleton. Their fins are supported by a fan of jointed rods called rays, made of bone or cartilage.

Ray-finned fish are able to swim with far greater precision than cartilaginous fish. Using their highly mobile and versatile fins, they can execute manoeuvres such as hovering, braking, and even swimming backwards. The fins themselves can be delicate and flexible or strong and spiny and they often have important secondary uses, such as in defence, display, and camouflage.

With the exception of many bottom-living and deep-sea species, most ray-finned fish have a buoyancy aid in the form of a gas-filled swim bladder. This allows them to adjust their buoyancy at different depths by adding or removing gas to and from the bladder via the bloodstream.

MYRIAD ADAPTATIONS

The great majority of fish are ray-finned and the group is hugely diverse – ranging from tiny gobies to the gigantic ocean sunfish. Species have evolved to inhabit every conceivable aquatic niche from tropical

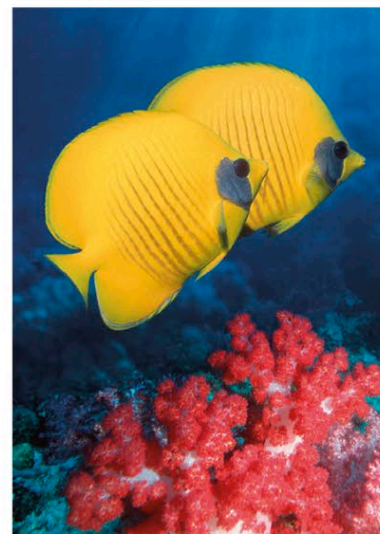
coral reefs to the waters beneath the ice-shelves of Antarctica, from the depths of the ocean to shallow desert pools. Herbivores, carnivores, and scavengers are all represented in the group and its members exhibit many ingenious hunting and defence strategies, as well as co-operation between species.

SAFETY IN NUMBERS

Most ray-finned fish shed eggs and sperm into the water and fertilization is external. In some cases, fewer eggs are laid and parental care is provided. For example, jawfish and some cichlids protect their eggs and young in their mouth while sticklebacks and many wrasses build nests of weed and debris. Some species protect their eggs with such vigour that even scuba divers are warned off.

Most species, however, lay eggs in vast numbers. The millions of floating eggs and fish larvae are an important food source for other aquatic creatures, but those that survive drift and disperse the species. Populations of fish reproducing in this way are less vulnerable to over-fishing as numbers can recover when fishing stops, but stocks – even of such prolific breeders as Atlantic cod – will eventually succumb if intense levels of fishing are continued.

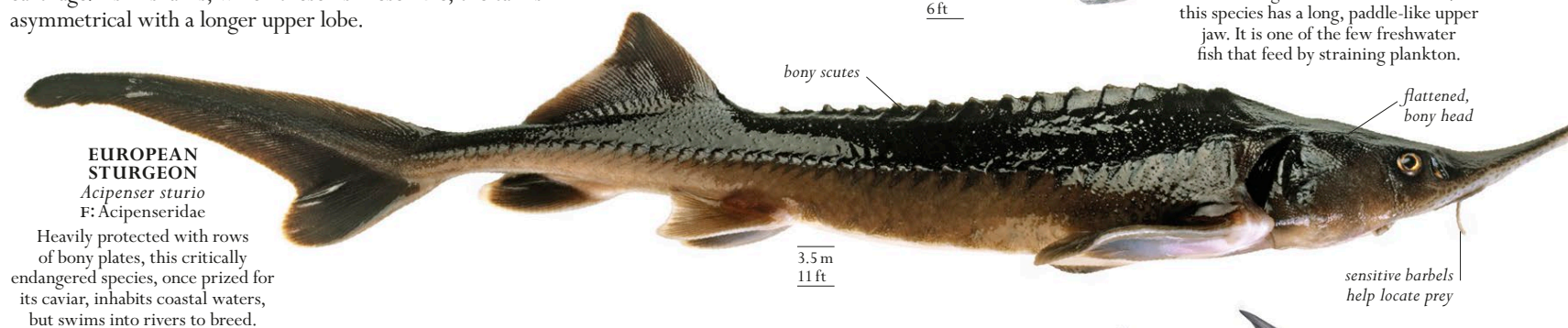
PHYLUM	CHORDATA
CLASS	ACTINOPTERYGII
ORDERS	47–59
FAMILIES	495–523
SPECIES	About 32,400



A pair of bluecheek butterflyfish (*Chaetodon semilarvatus*) swim in unison as they patrol their patch of coral reef.

STURGEONS AND RELATIVES

Among the 28 members of the order Acipenseriformes, only the sturgeon family includes marine species. The skull and some fin supports are made of hard bone, but most of the remaining skeleton is made of pliable cartilage. As in sharks, which these fish resemble, the tail is asymmetrical with a longer upper lobe.



EUROPEAN STURGEON
Acipenser sturio
F: Acipenseridae

Heavily protected with rows of bony plates, this critically endangered species, once prized for its caviar, inhabits coastal waters, but swims into rivers to breed.

AMERICAN PADDLEFISH
Polyodon spathula
F: Polyodontidae

Inhabiting N. American wetlands, this species has a long, paddle-like upper jaw. It is one of the few freshwater fish that feed by straining plankton.

GARS

Fish in the order Lepisosteiformes are primitive, freshwater predators from North America, with a long, cylindrical body protected by heavy, close-fitting scales.

Their long jaws have needle-like teeth.



LONGNOSE GAR
Lepisosteus osseus
F: Lepisosteidae

A skilled predator, this long, thin fish hangs motionless in the water, hidden by vegetation, before thrusting forwards to capture its prey.

TARPONS AND TENPOUNDERS

The order Elopiformes is small and its members are silvery with only one dorsal fin and a forked tail. They resemble giant herring. They have special throat bones (gular plates). Though these are marine fish, some species also swim up estuaries and rivers.

LADYFISH
Elops saurus
F: Elopidae

The ladyfish moves in large schools, close to W. Atlantic shores. When alarmed, it skips over the water's surface.



TARPON
Megalops atlanticus
F: Megalopidae

Found along Atlantic coasts, the tarpon sometimes enters rivers. In stagnant water, it gulps air from the surface using its swim bladder as a primitive lung.